

1. Project Title

YouTube Comment Sentiment & Trend Analyzer using Natural Language Processing (NLP)

2. Abstract

With the rapid growth of YouTube as one of the world's largest video-sharing platforms, millions of users express their opinions through comments. Analyzing these comments manually is time-consuming and inefficient. This project proposes an NLP-based system that automatically collects YouTube comments and analyzes them to determine audience sentiment, identify trending topics, extract important keywords, and generate concise summaries.

The system uses Natural Language Processing techniques such as text preprocessing, sentiment analysis, keyword extraction, topic modeling, and text summarization to provide valuable insights into viewer opinions. The results are displayed through an interactive dashboard, enabling content creators, businesses, and researchers to understand audience feedback quickly and make informed decisions.

3. Introduction

YouTube has become a major platform for education, entertainment, product reviews, marketing, and communication. Every popular video receives hundreds or thousands of comments that reflect audience opinions, suggestions, appreciation, or criticism.

Since manually analyzing these comments is impractical, Natural Language Processing (NLP) offers an efficient solution for automatically understanding and organizing textual feedback.

This project aims to build an intelligent YouTube Comment Analyzer capable of processing comments from any public YouTube video and presenting meaningful insights using AI techniques.

4. Problem Statement

YouTube videos often receive thousands of comments, making it difficult for creators and businesses to manually understand audience opinions and feedback.

The proposed system aims to develop an NLP-based application that automatically analyzes YouTube comments to determine viewer sentiment, identify trending topics, extract important keywords, and generate an overall summary of audience feedback, enabling faster and data-driven decision-making.

5. Objectives

The objectives of this project are:

- Fetch comments from a YouTube video using the YouTube Data API.
 - Clean and preprocess comment text.
 - Classify comments as Positive, Negative, or Neutral.
 - Extract frequently occurring keywords.
 - Identify major discussion topics.
 - Generate a concise summary of audience opinions.
 - Display analytical results through an interactive dashboard.
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6. Scope of the Project

The system is designed for analyzing comments from **public YouTube videos**.

It can be used by:

- YouTube content creators
- Marketing companies
- Product review analysts
- Researchers
- Educational content creators
- Digital media agencies

The project focuses on English-language comments (multilingual support can be added later).

7. Literature Review

Traditional methods of analyzing customer feedback relied on manual reading and categorization, which becomes impractical when dealing with thousands of

comments.

Recent advancements in Natural Language Processing (NLP) and transformer-based language models enable automatic sentiment detection, topic extraction, and text summarization. Techniques such as BERT, RoBERTa, and Sentence Transformers have significantly improved the accuracy of sentiment analysis and semantic understanding.

By combining these techniques with the YouTube Data API, it is possible to build an intelligent system that provides meaningful insights from audience comments in real time.

8. Proposed System

The proposed system consists of six major modules:

1. Comment Collection
 2. Text Preprocessing
 3. Sentiment Analysis
 4. Keyword & Topic Extraction
 5. Text Summarization
 6. Dashboard Visualization
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9. System Architecture

User

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Enter YouTube URL

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YouTube Data API

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Fetch Comments

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Text Preprocessing

- ├— Remove URLs
- ├— Remove Emojis
- ├— Tokenization
- ├— Stop-word Removal
- └— Lemmatization

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NLP Processing

- ├— Sentiment Analysis
- ├— Keyword Extraction

└─ Topic Detection
└─ Summarization

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Analytics Dashboard

10. Methodology

Step 1: Input

The user enters a public YouTube video URL.

Example:

```
https://www.youtube.com/watch?v=XXXXXXXXX
```

Step 2: Fetch Comments

The application retrieves comments using the YouTube Data API.

Collected information includes:

- Comment text
- Username (optional)
- Number of likes
- Date and time (optional)

Step 3: Text Preprocessing

The fetched comments undergo preprocessing.

Operations include:

- Convert text to lowercase
- Remove punctuation
- Remove URLs
- Remove emojis
- Remove stop words
- Tokenization

- Lemmatization

Example

Original:

```
OMG!!! 🥰🥰  
  
This tutorial is soooo good!!  
  
https://abc.com
```

Processed:

```
tutorial good
```

Step 4: Sentiment Analysis

Each comment is classified as:

- Positive
- Negative
- Neutral

Example

Comment	Sentiment
Excellent tutorial	Positive
Audio quality is bad	Negative
Nice video	Positive
Uploaded today	Neutral

Step 5: Keyword Extraction

Frequently occurring words are identified.

Example:

Python
Tutorial
AI
Camera
Editing

Step 6: Topic Detection

Comments are grouped into themes.

Example:

Topic 1

Audio Quality

Topic 2

Video Editing

Topic 3

Content Quality

Step 7: Text Summarization

Generate a short summary.

Example:

Most viewers appreciated the explanation and presentation style. Several users suggested improving audio quality and requested more advanced tutorials.

Step 8: Dashboard

Display:

- Total Comments

- Positive Comments
 - Negative Comments
 - Neutral Comments
 - Top Keywords
 - Topic Clusters
 - Overall Summary
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11. NLP Techniques Used

- Text Cleaning
 - Tokenization
 - Stop-word Removal
 - Lemmatization
 - Sentiment Analysis
 - Keyword Extraction
 - Topic Modeling
 - Text Summarization
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12. Technologies Used

Component	Technology
Programming Language	Python
IDE	VS Code / PyCharm
NLP Libraries	NLTK, spaCy
Machine Learning	Scikit-learn
Deep Learning	Hugging Face Transformers (optional)
API	YouTube Data API v3
Dashboard	Streamlit

Data Visualization	Plotly, Matplotlib, Seaborn
Database	SQLite (Optional)

13. Software Requirements

- Python 3.11+
 - Streamlit
 - pandas
 - numpy
 - nltk
 - spaCy
 - scikit-learn
 - matplotlib
 - plotly
 - google-api-python-client
 - transformers (optional)
-

15. Expected Output

After analyzing approximately 5000 comments:

```
Total Comments : 5000
```

```
Positive : 71%
```

```
Negative : 18%
```

```
Neutral : 11%
```

Top Keywords

```
AI
```

```
Tutorial
```

```
Camera
```

Editing

Python

Detected Topics

Content Quality

Audio

Future Tutorials

Summary

Most viewers appreciated the educational content and presentation quality. The primary criticism was related to audio quality, while many viewers requested additional advanced tutorials.

16. Advantages

- Saves time by automating comment analysis.
- Provides instant audience insights.
- Helps improve future video content.
- Supports marketing and product analysis.
- Easy-to-use dashboard.
- Scalable to large numbers of comments.

17. Limitations

- Requires access to public YouTube comments.
- Performance depends on the quality of the sentiment model.
- Initially supports English comments only.
- API quotas may limit the number of requests.

18. Future Enhancements

- Multilingual sentiment analysis.
- Emotion detection (Joy, Anger, Sadness, etc.).
- Spam and bot comment detection.
- Real-time monitoring of live stream comments.

- Comparative analysis between multiple YouTube videos.
 - AI-powered reply suggestions for creators.
 - Trend analysis over time.
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19. Applications

- YouTube Channel Analytics
 - Product Review Analysis
 - Brand Reputation Monitoring
 - Marketing Campaign Evaluation
 - Educational Content Feedback
 - Public Opinion Research
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20. Conclusion

The **YouTube Comment Sentiment & Trend Analyzer using NLP** provides an intelligent solution for understanding audience feedback at scale. By combining comment retrieval through the YouTube Data API with NLP techniques such as sentiment analysis, keyword extraction, topic detection, and summarization, the system transforms unstructured comments into meaningful insights.

The project demonstrates practical applications of Natural Language Processing while addressing a real-world problem faced by content creators, businesses, and researchers. It also provides a solid foundation for future enhancements such as multilingual support, emotion analysis, and AI-driven recommendation systems, making it a valuable academic mini project as well as a portfolio piece.